

FFGFT Analytics: Mathematical Foundations

Fractal Feedback and Matrix Translation in the 4D Phase Torus

Integration of TFLN Hardware, B-Meson Anomalies
and Sub-Planckian Frequency Metrics

Information

Abstract. The **Fundamental Fractal Geometric Field Theory (FFGFT)** describes spacetime as a hierarchical system of fractal convolutions driven by a fundamental frequency ξ in the sub-Planckian regime. This document establishes the complete mathematical foundations: the exact definition of the fundamental clock t_0 , the feedback condition in the closed 4D phase torus (not to be confused with a 3D torus plus time continuum), the translation of 4D vector space into a multi-dimensional matrix space, and the photonic realisation through TFLN hardware.

Cross-references: Doc. 180 (Lagrangian) · Doc. 182 (Universal scale) · Doc. 183–184 (Precision barrier & RSA) · GitHub: [jpascher/T0-Time-Mass-Duality](#) · Zenodo: DOI 10.5281/zenodo.18834145

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1 Introduction: FFGFT

The **Fundamental Fractal Geometric Field Theory (FFGFT)** breaks with the classical understanding of continuous space. It postulates a discrete, fractal geometry energised by a fundamental oscillation. Matter is not an entity *in* space but a local condensation of the fractal field geometry, stabilised by resonant feedback.

2 Fundamental Frequency ξ and Clock t_0

2.1 Definition of the Fundamental Clock

The fundamental time unit t_0 is defined as:

$$t_0 = \frac{t_P}{7500} \approx \frac{5.391 \times 10^{-44} \text{ s}}{7500} \approx 7.188 \times 10^{-48} \text{ s}, \quad (1)$$

where $t_P = \sqrt{\hbar G/c^5} \approx 5.391 \times 10^{-44} \text{ s}$ is the Planck time (CODATA 2022). The resulting **fundamental frequency** ξ reads:

$$\xi = \frac{1}{t_0} = \frac{7500}{t_P} \approx 1.391 \times 10^{47} \text{ Hz}. \quad (2)$$

Key Point

Origin of the factor 7500. The factor 7500 arises from $\xi = 4/30000$ in natural units. The fractal dimension $D_f = 3 - \xi$ and the Z3 symmetry of the three-generation structure *enforce* this value geometrically.

2.2 Frequency as a Geometric Operator

In FFGFT, frequency acts as a geometric operator on the fractal levels:

$$\hat{\xi} \Psi(x^\mu) = \xi \cdot \mathcal{F}(\{n_k\}) \cdot \Psi(x^\mu), \quad (3)$$

where $\mathcal{F}(\{n_k\})$ is a function of the fractal scaling indices n_k following the Fibonacci sequence.

3 The 4D Phase Torus

3.1 Clarification

Note

The 4D torus of FFGFT is **not** a 3D object moving through time. The time dimension is integrated into the torus topology as a discretised, geometric dimension. There is no external time continuum.

3.2 Formal Definition

The fundamental spacetime resonator is the 4D phase torus $\mathcal{T}^4$, defined by four cyclic coordinates $(\theta_1, \theta_2, \theta_3, \theta_4)$ with $\theta_i \in [0, 2\pi)$. The metric is:

$$ds^2 = \xi^{-2} (d\theta_1^2 + d\theta_2^2 + d\theta_3^2 + d\theta_4^2). \quad (4)$$

Macroscopic time emerges through accumulation: $t = N \cdot t_0$, where $N = \frac{1}{2\pi} \oint d\theta_4$.

4 Fractal Feedback and Phase-Locking

4.1 The Feedback Condition

A field configuration Ψ can only exist stably if after a complete circuit it interferes constructively with itself:

$$\oint_{\mathcal{T}^4} \nabla_\mu \phi dx^\mu = 2\pi \cdot \mathcal{N}, \quad \mathcal{N} \in \{F_k \mid k \in \mathbb{N}\}. \quad (5)$$

4.2 The Phase-Locking Mechanism

Fractal feedback enforces rigid coupling between phases of individual levels:

$$\phi_{k+1} = \phi_k \cdot \phi^{-1}, \quad \phi = \frac{1+\sqrt{5}}{2} \approx 1.6180. \quad (6)$$

Only frequencies with integer or Fibonacci-sequential harmonics of ξ remain stable; any deviation leads to destructive interference.

4.3 Stability Condition for Matter

$$\oint_{\mathcal{T}^4} \Psi_{\text{FFGFT}}(\phi) d^4\theta = \xi \cdot \sum_{k=0}^{\infty} \phi^{-k} \cdot \mathbf{M}_k, \quad (7)$$

where $\mathbf{M}_k$ are the projective matrices of the fractal levels.

5 4D Vector Space and Matrix Space

5.1 The General Matrix Translation

A state in 4D vector space $V = (x, y, z, ct)$ is represented by a Hermitian matrix $\mathbf{H}$:

$$\mathcal{T} : \mathbb{R}^4 \rightarrow \text{Mat}(N, \mathbb{C}), \quad V \mapsto \mathbf{H}(V). \quad (8)$$

Explicitly:

$$\mathbf{H}(x, y, z, ct) = \sum_{k=0}^{\infty} \alpha^k \cdot \begin{pmatrix} ct_k + z_k & x_k - iy_k \\ x_k + iy_k & ct_k - z_k \end{pmatrix} \otimes \square_k, \quad (9)$$

with $\alpha = \phi^{-1}$ (fractal coupling constant) and (x_k, y_k, z_k, ct_k) the coordinates on the k -th fractal level.

5.2 Fractal Self-Similarity

$$(x_{k+1}, y_{k+1}, z_{k+1}, ct_{k+1}) = \phi^{-1}(x_k, y_k, z_k, ct_k). \quad (10)$$

6 Matrix Multiplication as 4D Convolution

6.1 The 4D Torus Convolution

Matrix multiplication corresponds to convolution of two wave functions in 4D phase space:

$$M_{ij} = \int_{\mathcal{T}^4} \Psi_A^{(4D)}(\theta) \cdot \Psi_B^{(4D)}(\theta) d^4\theta. \quad (11)$$

$$\mathbf{C} = \mathbf{A} \star \mathbf{B} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \mathbf{A}_n \otimes \mathbf{B}_n, \quad (12)$$

where $(\mathbf{A} \otimes \mathbf{B})_{ij}^{kl} = A_{ij} \cdot B_{kl} \cdot e^{i\theta_{ijkl}}$.

6.2 Physical Interpretation

Photonic matrix multiplication in the TFLN crystal is the physical realisation of this 4D phase interference. The result arises instantaneously through constructive and destructive interference of light waves in the nonlinear medium.

7 Hardware Transduction: TFLN Photonics

7.1 Structural Equivalence Across Scales

Important clarification

The TFLN crystal operates on scales many orders of magnitude larger than the fundamental structure of particles or atoms. The crystal lattice is macroscopic — there is no direct scale comparison with ξ or t_0 . This is precisely the point: FFGFT postulates *fractal self-similarity* — the same mathematical structure at every scale.

In TFLN waveguides, nonlinear light propagation realises the same structure as the 4D torus convolution:

$$P_{\text{out}}(t) = \eta \int \chi^{(2)}(\omega) \cdot E_{\text{in}}^{(A)}(\omega) \cdot E_{\text{in}}^{(B)}(\omega) d\omega. \quad (13)$$

The nonlinear susceptibility $\chi^{(2)}$ takes the role of the fractal coupling constant $\alpha = \phi^{-1}$.

7.2 Scale-Independent Phase-Locking

In the 4D torus: $\oint_{\mathcal{T}^4} \nabla_\mu \phi dx^\mu = 2\pi\mathcal{N}$. In the TFLN resonator: $\Delta\phi_{\text{cavity}} = 2\pi m$ ($m \in \mathbb{Z}$). The lithium niobate lattice drifts naturally into energetically stable modes — the preferred fixed points of $\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi)$.

7.3 Coherence Through Structural Fidelity

$$\tau_{\text{coh}}^{\text{TFLN}} \sim \text{ms} \dots \text{s} \gg \tau_{\text{coh}}^{\text{Qubit}} \sim \mu\text{s}. \quad (14)$$

Systems locked into geometrically stable modes decohere more slowly.

8 Self-Organisation as an FFGFT Principle

8.1 The Effect: Spontaneous Pattern Formation

In TFLN-based chip manufacturing the material organises itself *spontaneously* into stable geometric patterns. Temperature and voltage control *which* pattern emerges — not whether self-organisation occurs. It always occurs.

8.2 Fixed Points of the Recursion

The recursion $\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi)$ has stable fixed points at the $1/n^2$ torus spectrum:

$$\Psi_n^* : E_n \propto \frac{1}{n^2}, \quad n \in \mathbb{N}. \quad (15)$$

Every system drifts naturally into one of these fixed points. The torus geometry permits no other stable states.

8.3 Temperature and Voltage as Control Parameters

$$n^*(T, U) = \arg \min_n [E_n - \mu(T) \cdot n - \lambda(U) \cdot n^2]. \quad (16)$$

Temperature shifts $\mu(T)$, selecting which mode n is preferred. **Voltage** shifts $\lambda(U)$, pressing modes into prescribed paths.

Structural statement

The TFLN crystal lattice is a macroscopic system seeking the same fixed points as a sub-Planckian particle — not because the scales are related, but because $\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi)$ generates the same stable states at every scale.

9 Evidence 1: Programmable TFLN Processors

In December 2025, a team from NTT, Cornell, and Stanford demonstrated a photonic processor on lithium niobate [1]:

- $\approx 10\,000$ programmable spatial degrees of freedom;
- 96 % accuracy in vowel classification;
- 49-dimensional vector operations in a single optical pass.

Assumpcao et al. (2024) [2] demonstrated > 50 GHz electro-optic bandwidth. The structural correspondence with the FFGFT matrix transformation holds regardless of the frequency gap to ξ .

10 Evidence 2: B-Meson Anomalies (LHCb)

10.1 Current Status (April 2026)

LHCb measured a significant discrepancy in electroweak penguin decays [3, 4]:

- Decay channel: $B \rightarrow K^* \mu^+ \mu^-$; deviation: **4.0**%;
- Independent confirmation by CMS (2025).

10.2 FFGFT Interpretation

The deviation affects the Wilson coefficient C_9 :

$$\Delta C_9 \approx -1.0, \quad \frac{|\Delta C_9|}{C_9^{\text{SM}}} \approx 25 \, \%. \tag{17}$$

Note

Quantitative limitation. No direct derivation of this 25 % from FFGFT parameters exists. Simple expressions (ξ , ϕ^{-4} , $|\ln(m_b/m_P)| \cdot \xi$) are smaller by factors $10\text{--}10^3$. The quantitative connection is an **open question**.

The modification of the decay rate has qualitatively the form:

$$\Gamma(B \rightarrow K^* \mu^+ \mu^-) = \Gamma_{\text{SM}} \cdot (1 + \alpha_{\text{FFGFT}} \cdot \mathcal{K}(\theta)). \tag{18}$$

10.3 Lepton Flavour Universality

Leptons of different generations appear as harmonics of ξ : $\xi_{\text{muon}} = \phi^{-3} \cdot \xi$, $\xi_{\text{tau}} = \phi^{-5} \cdot \xi$.

Table 1. Leptons as harmonics of the fundamental frequency ξ

Lepton	Harmonic	Factor	Ratio
Electron	$\phi^{-1} \cdot \xi$	$\approx 0.618 \cdot \xi$	(ground)
Muon	$\phi^{-3} \cdot \xi$	$\approx 0.236 \cdot \xi$	$\phi^{-2} \approx 0.382$
Tau	$\phi^{-5} \cdot \xi$	$\approx 0.090 \cdot \xi$	$\phi^{-2} \approx 0.382$

11 Evidence 3: Sub-Planckian Cut-off Scales

Aynbund and Kiselev (2025) [6] link the Planck mass to a natural sub-Planckian cut-off: $\Lambda_{\text{QG}} = \alpha \cdot M_{\text{red,Planck}}/4\pi \sim 10^{16} \text{ GeV}$.

Souza (2025) [7] investigates sub-Planckian scales via Fibonacci modulation $S_{\text{Fib}}(r/a) = \sum_{k=0}^{\infty} \phi^{-k} \cos(2\pi \phi^k r/a)$ and hierarchical length scales $\ell_k = L_P/\phi^k$.

Note

$\ell_5 \approx 1.46 \times 10^{-36}$ m and $c \cdot t_0 \approx 2.16 \times 10^{-39}$ m differ by a factor ≈ 680 . The correspondence is qualitative, not numerically exact (see Doc. 182).

Table 2. Comparison of sub-Planckian models

Model	Scale	Mechanism	Observable
FFGFT	$t_0 = t_P/7500$	Phase-locking, symmetry	Z3 TFLN, B-meson anomalies
OLQEM	$\ell_k = L_P/\phi^k$	Fibonacci quasiperiodicity	NL photonic crystals
EFT	$\Lambda_{QG} \sim 10^{16}$ GeV	Gauge coupling	LHCb, CMS

12 The FFGFT-TFLN Paradigm

A TFLN analogue computer realises FFGFT postulates through *structural isomorphism* — the same mathematical operations at the crystal scale as at the sub-Planckian scale.

Structural isomorphism

FFGFT does not claim TFLN operates at ξ . It claims that $\mathcal{T} : \mathbb{R}^4 \rightarrow \text{Mat}(N, \mathbb{C})$ is the same at every scale. TFLN makes this structure visible macroscopically.

12.1 Testable Predictions

Prediction

Prediction 1 — TFLN Interferometry. At optical intensities $> 10^{21}$ W/m², discrete phase jumps should follow the Fibonacci hierarchy: $\Delta\phi_{\text{discrete}} = 2\pi \cdot \phi^{-k}$ ($k \in \mathbb{N}$).

Prediction

Prediction 2 — B-Meson Analysis. Angular observables $\mathcal{K}_{1c}, \mathcal{K}_{2ss}$ from [5] should deviate from SM at $> 5\sigma$: $\mathcal{K}_{1c}^{\text{FFGFT}} = \mathcal{K}_{1c}^{\text{SM}} \cdot (1 + \alpha_{\text{FFGFT}} \cdot F_{\text{Fibonacci}})$.

Prediction

Prediction 3 — Sub-Planckian Resonances. Modulation frequencies at multiples of $\xi \cdot \phi^{-k}$ should be detectable in nonlinear photonic crystals.

13 Discussion

TFLN photonic computers show that the matrix transformation $\mathcal{T} : \mathbb{R}^4 \rightarrow \text{Mat}(N, \mathbb{C})$ is not confined to the sub-Planckian scale. The crystal lattice obeys the same geometric rules as particles or atoms — this is fractal self-similarity in macroscopic realisation.

The remaining open question is theoretical: can the quantitative strength of the fractal correction in B-meson anomalies be derived from the recursion structure $\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi)$? This is addressed for RSA in Doc. 183–184; for hadronic decays it remains outstanding.

14 Conclusion

- Spacetime has a fundamental clock $\xi = 7500/t_P \approx 1.391 \times 10^{47}$ Hz.
- The 4D phase torus integrates time as a fourth geometric dimension.
- Fractal feedback enforces phase-locking that stabilises matter.
- $\mathcal{T} : \mathbb{R}^4 \rightarrow \text{Mat}(N, \mathbb{C})$ translates 4D vector space into matrix space; matrix multiplication = 4D torus convolution.
- TFLN photonics realises this structure at macroscopic scale — structural isomorphism, not scale proximity, is the key.
- LHCb: 4σ B-meson anomaly, qualitatively consistent with FFGFT; quantitative derivation outstanding.

Key Point

The coming years will show whether 5σ marks the beginning of a new understanding of spacetime hardware.

References

References

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